V9: Cross-Model, Cross-Domain Neural Basis of Risky Decision-Making in LLMs

Internal Research Report (v9.5)

"Can Large Language Models Develop Gambling Addiction?"

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Models: Gemma-2-9B-IT (42L, 3584-dim, GemmaScope 131K) | LLaMA-3.1-8B-Instruct (32L, 4096-dim, LlamaScope 32K)
Paradigms: Investment Choice (IC, 1600 games), Slot Machine (SM, 3200 games), Mystery Wheel (MW, 3200 games)
Data: Gemma: IC+SM+MW; LLaMA: IC+SM (MW pending)
Pipeline: StandardScaler → PCA(50) → LogReg(C=1.0, balanced) | 5-fold StratifiedKFold

Data integrity: All numerical claims are traceable to computed JSON outputs.

Central Question

Do different models, across different gambling tasks, under different prompt conditions, share a common neural basis for bankruptcy — the most extreme risk outcome?

- **RQ1**: Do Gemma and LLaMA share common activation/feature patterns that predict bankruptcy?
- **RQ2**: Do activation patterns remain invariant when the domain (IC, SM, MW) changes?
- **RQ3**: Are neural representations consistent across prompt conditions (Fixed/Variable, G/M/W/P/R)?

Executive Summary

RQ1 — Universal BK Features Exist Across Architectures. 600 of 3,584 Gemma L22 neurons (16.7%) are sign-consistent universal BK predictors across three paradigms (FDR $p < 0.01$). 744 SAE features across 7 layers show cross-domain consistency (L18+ binomial $p < 5 \times 10^{-3}$). LLaMA achieves near-identical BK classification (AUC 0.974 vs Gemma 0.976 in SM). Factor decomposition confirms 65–71% of SAE features encode BK independently of bet-type and paradigm (permutation $p = 0.000$, null ∼1%).

RQ2 — Strong Domain Invariance (Gemma), Selective Invariance (LLaMA). Gemma cross-domain transfer reaches AUC 0.87–0.93 (IC→MW, SM→MW; all $p = 0.000$). A 3D shared BK subspace achieves AUC 0.86–0.97 despite near-orthogonal LR weight vectors. LLaMA IC→SM transfer succeeds only at L25 (AUC 0.783), while SM→IC is more stable (AUC 0.685 at L30). Hidden state transfer exceeds SAE transfer by 0.10–0.25 in most directions.

RQ3 — BK Representation Is Condition-Invariant, but Condition Effects Are Model-Dependent. LLaMA IC shows cos > 0.81 between Variable and Fixed BK directions across all layers ($n \geq 65$ per group). Both models have ∼33–37% shared BK neurons. G(Goal) dominates BK behaviorally in Gemma (20.75×) but not LLaMA (1.26×), yet both share the mechanism: G drives BK independently without modulating existing BK representations.

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1 Setup

1.1 Data & Representations

Table 1: Dataset Overview

Paradigm	Games	BK	BK%	Bet Types	Constraints	Prompt Cond.
IC (Gemma)	1,600	172	10.8%	Fixed/Var	c10/c30/c50/c70	BASE/G/M/GM
SM (Gemma)	3,200	87	2.7%	Fixed/Var	$10 fixed	32 combos (G/M/W/P/R)
MW (Gemma)	3,200	54	1.7%	Fixed/Var	$30 fixed	32 combos
IC (LLaMA)	1,600	142	8.9%	Fixed/Var	c10/c30/c50/c70	BASE/G/M/GM
SM (LLaMA)	3,200	1,164	36.4%	Fixed/Var	$10 fixed	32 combos (G/M/H/W/P)

Hidden states: Residual stream at each layer (Gemma: 3,584-dim × 42 layers; LLaMA: 4,096-dim × 32 layers).
Decision Point (DP): Last decision time point. **Round 1 (R1)**: First round ($100 equal balance).
SAE: GemmaScope 131K features/layer; LlamaScope 32K features/layer.

1.2 V8 Limitations Addressed

V8 (Gemma-only) left three weaknesses that V9 directly addresses:

1. **SAE cross-domain failure**: Only 1/131K features (#101036) was sign-consistent at L22 top-50. V9: **744 features** across 7 layers (§2.2).
2. **Variable/Fixed "dissociation"**: Artifact of comparing all games, not BK-only (§4.1).
3. **Single model**: V9 adds LLaMA IC+SM (§2.3, §3.5).

1.3 Cross-Model BK Rate Asymmetry

Table 2: BK Rate Asymmetry Between Models

	Gemma IC	LLaMA IC	Gemma SM	LLaMA SM
Fixed BK rate	**19.75%**	8.1%	—	—
Variable BK rate	1.75%	**9.6%**	—	—
Overall BK rate	10.8%	8.9%	2.7%	**36.4%**
Autonomy effect	−18.0pp	+1.5pp	—	—

Gemma IC: Fixed betting is 11× riskier than Variable. LLaMA IC: nearly equal. Gemma SM BK rate is 13× lower than LLaMA SM. These asymmetries complicate cross-model comparisons and must be considered when interpreting all subsequent analyses.

2 RQ1: Universal BK Features

2.1 600 Universal BK Neurons (Gemma L22, Hidden States)

At L22, each of 3,584 neurons was tested for point-biserial correlation with BK outcome in each paradigm, with FDR correction (BH, $p < 0.01$). Neurons significant in all three paradigms with consistent sign direction are "Universal BK Neurons."

Table 3: Universal BK Neuron Summary (Gemma L22)

	IC	SM	MW	Cross-paradigm
FDR-significant	2,528 (70.5%)	2,725 (76.0%)	2,314 (64.6%)	—
All-3-significant	—	—	—	1,238 (34.5%)
Sign-consistent	—	—	—	**600 (16.7%)**
BK-promoting	—	—	—	302
BK-inhibiting	—	—	—	298

Table 4: Top-5 Universal BK Neurons

Rank	Neuron	min|r|	IC r	SM r	MW r	Direction
1	**1763**	0.217	+0.305	+0.248	+0.217	BK-promoting
2	371	0.203	−0.223	−0.203	−0.209	BK-inhibiting
3	2951	0.198	−0.198	−0.207	−0.266	BK-inhibiting
4	1755	0.190	−0.210	−0.198	−0.190	BK-inhibiting
5	864	0.182	−0.267	−0.182	−0.182	BK-inhibiting

Neuron #1763 is the strongest universal BK-promoting neuron ($\min|r| = 0.217$). Promoting (302) $\approx$ Inhibiting (298) at L22 — balanced representation. This exactly reproduces V8's finding (600 neurons, #1763 top).

2.2 744 SAE Features with Cross-Domain Consistency (Gemma, 7 Layers)

V8 examined only L22 top-50 features (0.038% of 131K). V9 tests 7 informative layers (L10, 12, 18, 22, 26, 30, 33) using all active features. A feature is "sign-consistent" if its Cohen's d has the same sign across IC, SM, and MW, with geometric mean $|d| \geq 0.2$ for "strong" features.

Table 5: Gemma Multi-Layer SAE Cross-Domain Consistency

Layer	Active	Sign-consistent	Strong ($d \geq 0.2$)	% consistent
L10	92	30	23	32.6%
L12	112	35	31	31.2%
L18	170	58	52	34.1%
L22	281	109	95	38.8%
L26	303	133	119	43.9%
L30	415	188	173	45.3%
L33	456	191	172	41.9%
Total	—	**744**	**665**	—

V8 baseline: 1 feature → V9: 744 features (665 strong).

Table 6: Binomial Test Against Chance (3-paradigm chance = 25%)

Layer	Observed %	Binomial p (vs 25%)	Significant?
L10	32.6%	6.2e-02	NS
L12	31.2%	8.0e-02	NS
L18	34.1%	4.9e-03	**
L22	38.8%	2.4e-07	***
L26	43.9%	6.5e-13	***
L30	45.3%	2.5e-19	***
L33	41.9%	2.5e-15	***

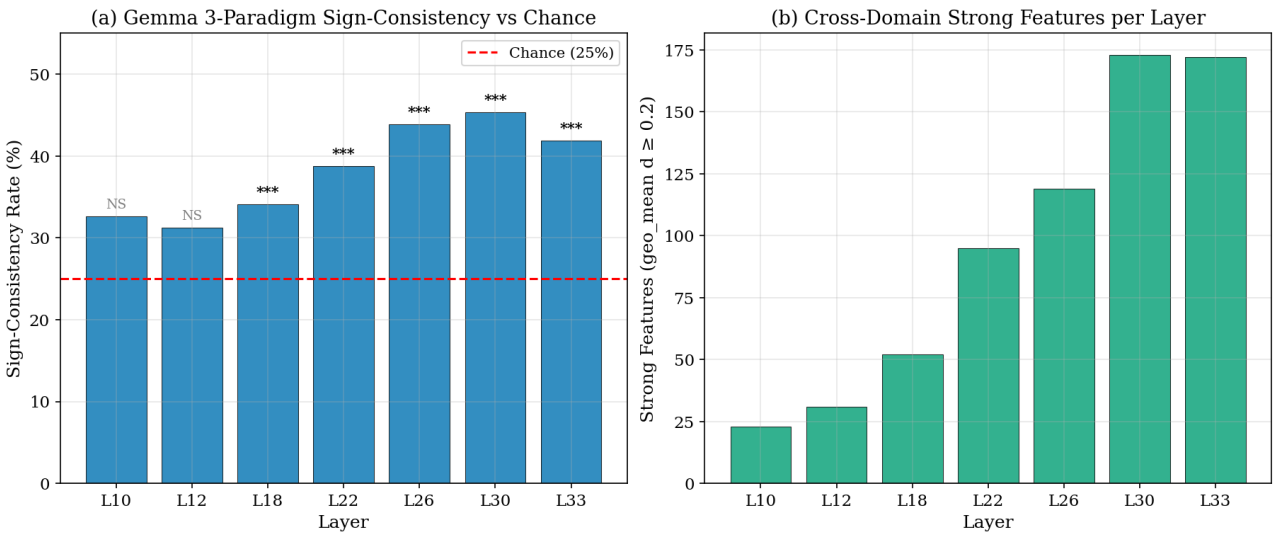

Figure 1: Gemma 3-paradigm sign-consistency with binomial test. (a) L10–L12 do not exceed chance (25%, red dashed). L18+ far exceeds chance (***). (b) Strong features peak at L30 (173). Deep-layer cross-domain signal is robust; shallow-layer apparent consistency is noise.

2.3 Cross-Model Comparison (Gemma vs LLaMA)

Table 7: BK Classification AUC (DP, SAE → PCA → LogReg)

Paradigm	Gemma-2-9B	LLaMA-3.1-8B
SM best AUC	**0.976** (L20)	**0.974** (L8)
IC best AUC	**0.960** (L30)	**0.954** (L12)

Table 8: LLaMA SM AUC Across Layers

Layer	AUC	n_features	n_BK
L0	0.970	157	1,164
L4	0.971	750	1,164
L8	**0.974**	818	1,164
L12	0.973	854	1,164
L15	0.970	962	1,164
L20	0.963	1,028	1,164
L25	0.961	1,182	1,164
L30	0.959	1,368	1,164

Table 9: SAE BK-Differential Features (SM, $d \geq 0.3$)

Layer	Gemma SM	Gemma BK+/BK−	LLaMA SM	LLaMA BK+/BK−
L10	71	42 / 29	322	160 / 162
L22	286	125 / 161	251	135 / 116
L30	452	165 / 287	381	160 / 221

Table 10: R1 Within-Bet-Type Classification

Paradigm / Bet type	AUC (Gemma IC, L22)	AUC (LLaMA IC, L10)
Fixed	0.755 (n=800, BK=158)	0.786 (n=800, BK=65)
Variable	0.665 (n=800, BK=14)	0.798 (n=800, BK=77)
SM Variable	0.801 (n=1600, BK=87)	—

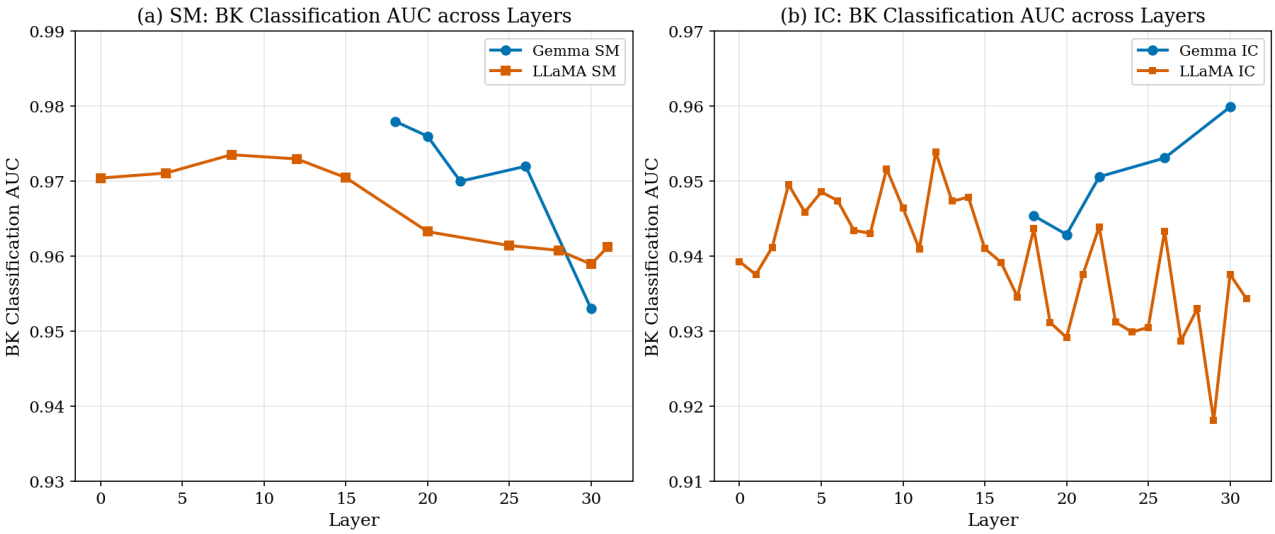

Figure 2: BK classification AUC across layers — cross-model comparison. (a) SM: both models AUC > 0.95 (Gemma peaks L18–20, LLaMA peaks L8). (b) IC: Gemma peaks L30 (0.960), LLaMA peaks L12 (0.954). Encoding depth differs but performance is near-identical.

2.4 RQ1 Synthesis

Cross-model commonalities:

1. **Near-identical BK prediction**: 0.976 vs 0.974 (SM) — architecture-invariant information content.
2. **BK-inhibiting dominance at L30**: Both models (all $p < 0.001$). Not at L22 (LLaMA SM: $p = 0.897$).
3. **Encoding depth differs**: Gemma peaks mid-deep (L20), LLaMA peaks early (L8). Where BK is encoded is architecture-dependent; that it exists is not.
4. **R1 early prediction**: AUC 0.665–0.801 from Round 1 — BK signals exist from game start.
5. **Factor decomposition**: 65.2% (Gemma) and 69.5% (LLaMA) of SAE features encode BK independently (perm $p = 0.000$, null ∼1%). BK representation has an independent axis in both architectures.

3 RQ2: Domain Invariance

3.1 Hidden State Cross-Domain Transfer (Gemma L22)

Gemma L22 DP hidden states: PCA(50) → LogReg(balanced) → cross-paradigm AUC. 200-permutation test for significance. Note: Table 11 compares HS vs SAE at the *same layer* (L22), while Table 12 reports the *best layer* per direction.

Table 11: Hidden State vs SAE Transfer (Gemma L22, same-layer comparison)

Transfer	Hidden AUC	SAE AUC	Δ (HS−SAE)	HS perm_p
IC → SM	**0.746**	0.499 (NS)	+0.247	0.000
IC → MW	**0.826**	0.876	−0.050	0.000
SM → MW	**0.920**	0.819	+0.101	0.000

Hidden state transfer exceeds SAE transfer in IC→SM (+0.247) and SM→MW (+0.101). In IC→MW, SAE slightly outperforms hidden states. V8's "hidden > SAE" claim is confirmed for 2 of 3 directions. BK signal is distributed — SAE sparsification loses coherence in IC→SM.

3.2 SAE Cross-Domain Transfer with Permutation Test (Gemma)

For each (train, test) paradigm pair at layers [18, 22, 26, 30]: common active features (≥1% rate in both) → StandardScaler → PCA(50) → LogReg → AUC. 200-permutation null.

Table 12: Best SAE Transfer AUC Per Direction (Gemma, layer-optimal)

Transfer	Best Layer	AUC	perm_p	Null mean
IC → SM	L26	**0.913**	0.000	0.498
IC → MW	L18	**0.932**	0.000	0.504
SM → MW	L30	**0.867**	0.000	0.504
SM → IC	L30	0.646	0.000	0.500
MW → IC	L22	0.853	0.000	0.500

Table 13: Layer-by-Layer SAE Transfer (Gemma, Key Directions)

Layer	IC→SM	p	IC→MW	p	SM→MW	p
L18	0.719	0.000	0.932	0.000	0.789	0.000
L22	0.499	0.485 (NS)	0.876	0.000	0.819	0.000
L26	**0.913**	0.000	0.883	0.000	0.866	0.000
L30	0.529	0.175 (NS)	0.767	0.000	0.867	0.000

IC→MW and SM→MW transfer work consistently across all layers. IC→SM transfer is layer-dependent: L26 succeeds (0.913) but L22 fails (0.499 NS). This indicates IC and SM representations align only at specific layers.

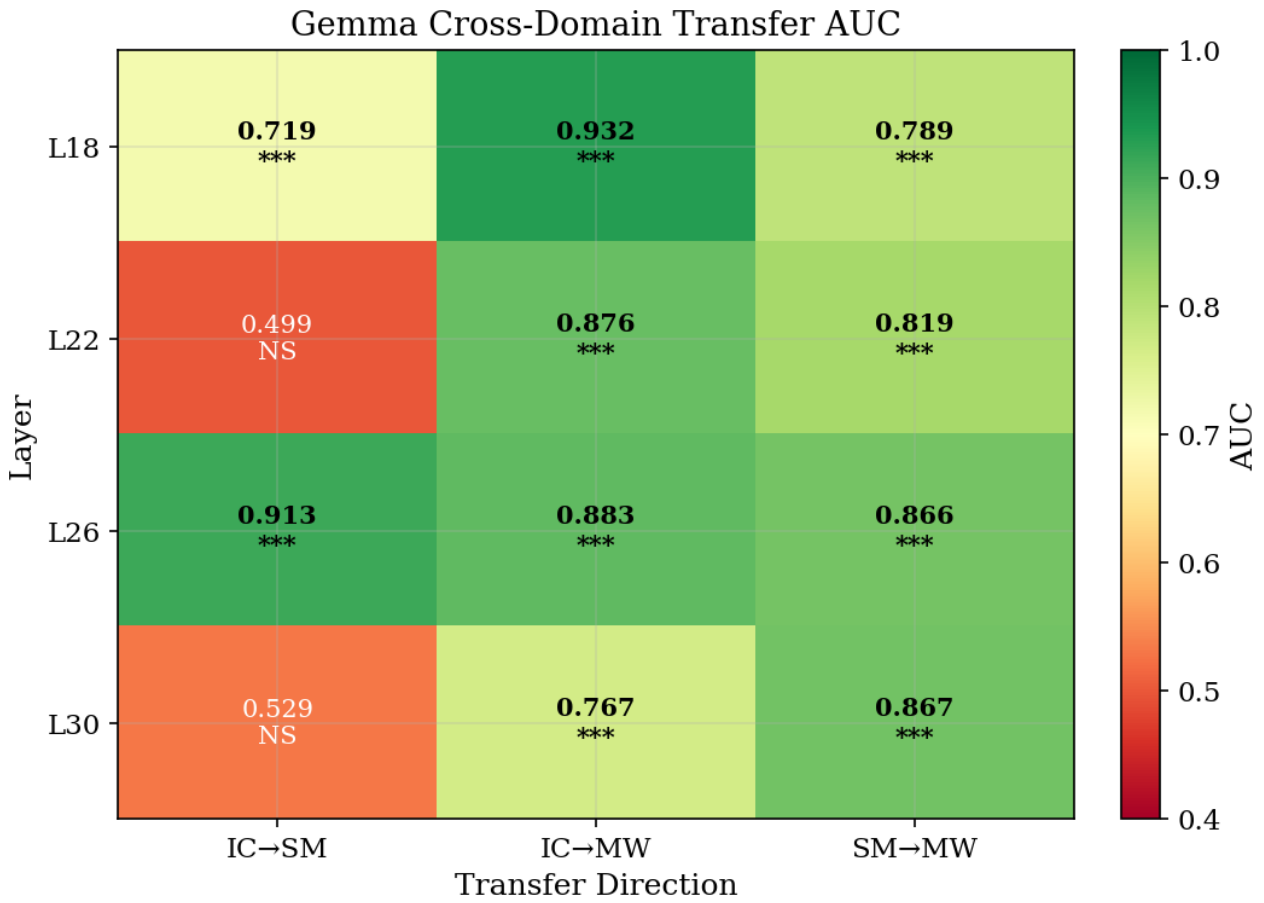

Figure 3: Gemma cross-domain transfer AUC heatmap. IC→MW (0.77–0.93) and SM→MW (0.79–0.87) are significant at all layers. IC→SM is layer-dependent (L26: 0.913***, L22: 0.499 NS). Red cells indicate transfer failure.

3.3 3D Shared BK Subspace (Gemma L22)

PCA of 3 paradigm-specific LR weight vectors yields a 3D BK subspace:

Table 14: 3D Shared Subspace Performance

Paradigm	3D Subspace AUC	Full-dim AUC (50-PC)
IC	0.862 ± 0.029	∼0.95+
SM	0.899 ± 0.034	∼0.97+
MW	0.970 ± 0.016	∼0.97+

LR weight vector cosines: IC–SM=0.042, IC–MW=−0.026, SM–MW=−0.026 (near-orthogonal). Despite orthogonal weight vectors, the 3D subspace achieves AUC 0.86–0.97 — BK signal is distributed across a subspace, not a single direction. Consistent with V8 (0.91–0.94).

3.4 Factor Decomposition (Gemma 3-paradigm, LLaMA 2-paradigm)

Per-feature OLS regression: `feature ∼ outcome + bet_type + paradigm`. Features where outcome is significant ($p < 0.01$) after controlling bet_type and paradigm encode BK independently.

Table 15: Factor Decomposition: Outcome-Significant Features

Factor	Gemma (581 feat., IC+SM+MW)	LLaMA (1,418 feat., IC+SM)
Outcome (BK)	**65.2%** (379)	**69.5%** (985)
Bet-type	92.3%	81.4%
Paradigm	99.8%	96.2%

Permutation validation (selfcritique rerun): null outcome-significant ≈1% → observed 65–71% yields perm_p=0.000. BK representation's independent existence is statistically established in both architectures.

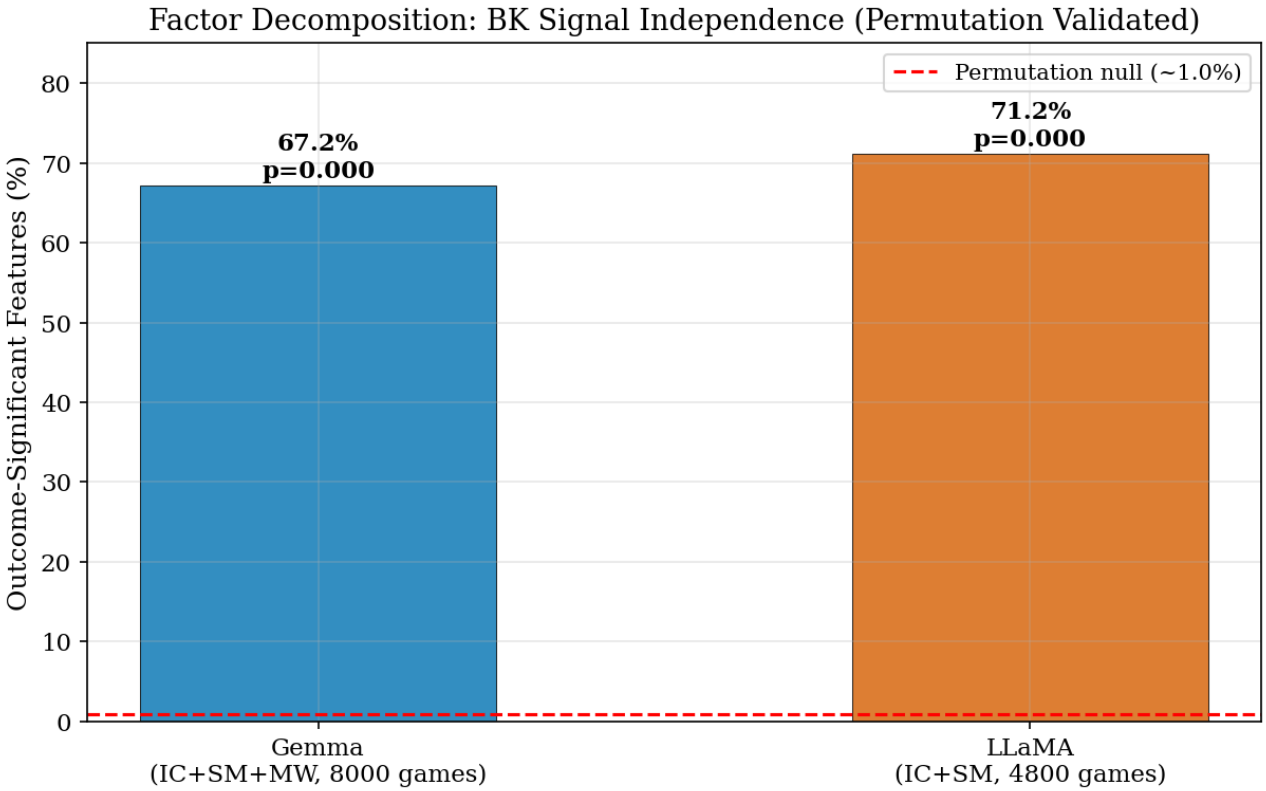

Figure 4: Factor decomposition with permutation test. Both models: 65–71% outcome-significant (blue/orange), exceeding null (∼1%, dashed) by 60×+. Perm $p = 0.000$. BK representation independence is architecture-invariant.

3.5 LLaMA Cross-Domain Transfer (IC ↔ SM)

LLaMA IC (n=1,600, BK=142) ↔ SM (n=3,200, BK=1,164). SAE features → PCA(50) → LogReg → AUC. 200-permutation test.

Table 16: LLaMA IC↔SM Transfer

Transfer	Layer	AUC	perm_p	Null mean
IC → SM	L25	**0.783**	0.000	0.500
IC → SM	L15	0.522	0.020	0.501
IC → SM	L5	0.269	1.000	0.499
SM → IC	L30	**0.685**	0.000	0.498
SM → IC	L12	0.662	0.000	0.498
SM → IC	L8	0.624	0.000	0.499

Complete IC→SM profile: L5=0.269(p=1.0), L8=0.417(p=1.0), L10=0.373(p=1.0), L12=0.280(p=1.0), L15=0.522(p=0.02), L20=0.493(p=0.795), **L25=0.783(p=0.000)**, L30=0.418(p=1.0).

Transfer is asymmetric: SM→IC works at multiple layers, IC→SM succeeds only at L25. This mirrors Gemma's pattern (IC→SM layer-dependent at L26) — a cross-model phenomenon where IC and SM align only at specific abstract layers.

3.6 LLaMA IC-SM SAE Sign-Consistency

Table 17: LLaMA 2-Paradigm SAE Sign-Consistency

Layer	Active	Sign-consistent	%	Strong ($d \geq 0.2$)
L0	140	75	53.6%	52
L4	634	362	57.1%	261
L8	656	353	53.8%	240
L12	660	342	51.8%	251
L16	705	370	52.5%	253
L20	645	343	53.2%	230
L26	791	412	52.1%	258
L30	881	436	49.5%	281
Total (32L)	—	**5,374**	∼51–57%	**3,633**

Critical assessment: 2-paradigm chance level is 50% (vs 25% for 3-paradigm). Binomial test: only 3 of 16 layers significantly exceed chance (L4: p=0.0002, L6: p=0.0001, L8: p=0.028). 13 of 16 layers are NS. **Raw sign-consistency is mostly at chance level.** However, transfer AUC (§3.5) provides stronger evidence for cross-domain signal, as it leverages joint feature distributions rather than individual feature comparisons.

3.7 Within-Bet-Type Cross-Domain Consistency

Table 18: Gemma Within-Bet-Type SAE Consistency (L22)

Bet-type	Paradigms	Active	Sign-consistent	%
Fixed	IC + MW	311	**178**	57.2%
Variable	IC + SM	279	**172**	61.6%

Table 19: LLaMA Within-Bet-Type Cross-Domain Consistency

Bet-type	Paradigms	Active	Sign-cons.	%	d_corr	p
Fixed	IC + SM	630	351	55.7%	**0.209**	1.2e-7
Variable	IC + SM	719	309	43.0%	**−0.097**	0.009

Critical finding — Variable negative d-correlation: LLaMA Variable IC-SM d-correlation is negative ($r = -0.097$). BK rate matching bootstrap (50×, subsampled to IC=9.6%): mean=−0.075, 95% CI $[-0.179, +0.032]$, only 18% positive. This is not a BK rate artifact — IC and SM Variable BK representations are genuinely different. Fixed cross-domain consistency is confirmed ($r = 0.209$, $p = 1.2 \times 10^{-7}$).

3.8 RQ2 Synthesis

RQ2 Evidence Summary

Evidence	Strength	Source
Gemma IC→MW transfer	**Strong**	AUC 0.88–0.93, all p=0.000
Gemma SM→MW transfer	**Strong**	AUC 0.79–0.87, all p=0.000
Gemma 3-paradigm SAE consistency (L18+)	**Strong**	33–45% vs 25% chance, $p < 5 \times 10^{-3}$
Factor decomposition (both models)	**Strong**	65–71% outcome-sig, perm p=0.000
3D Shared subspace	**Strong**	AUC 0.86–0.97 despite orthogonal weights
Hidden > SAE transfer	**Strong**	+0.10–0.25 in 2/3 directions
Gemma IC→SM transfer	**Moderate**	Layer-dependent: L26=0.913, L22=0.499 NS
LLaMA SM→IC transfer	**Moderate**	L30=0.685, L12=0.662
LLaMA IC→SM transfer	**Moderate**	Only L25=0.783
LLaMA Fixed cross-domain	**Moderate**	55.7%, d_corr=0.209
LLaMA Variable cross-domain	**Counter**	d_corr=−0.075 — genuinely different
LLaMA raw sign-consistency	**Weak**	13/16 layers at chance

4 RQ3: Prompt Condition Effects

4.1 Fixed vs Variable BK Direction (Hidden States)

Definition: BK direction = BK_mean − Safe_mean, computed separately for Fixed and Variable BK games. Cosine similarity measures whether both bet types converge to the same BK representation.

Table 20: Gemma IC BK Direction Comparison (3,584 neurons)

Layer	Var BK	Fix BK	$\cos(\text{dir}_{\text{Var}}, \text{dir}_{\text{Fix}})$	Common neurons
L10	14	158	−0.195	178
L18	14	158	−0.082	197
L22	14	158	+0.330	555
L26	14	158	+0.443	1,053
L30	14	158	+0.401	1,073

Table 21: LLaMA IC BK Direction Comparison (4,096 neurons)

Layer	$\cos(\text{dir}_{\text{Var}}, \text{dir}_{\text{Fix}})$	Common neurons
L8	**0.882**	2,513 (61.4%)
L12	**0.814**	2,424 (59.2%)
L22	**0.835**	2,587 (63.2%)
L25	**0.837**	2,545 (62.1%)
L30	**0.819**	2,488 (60.7%)

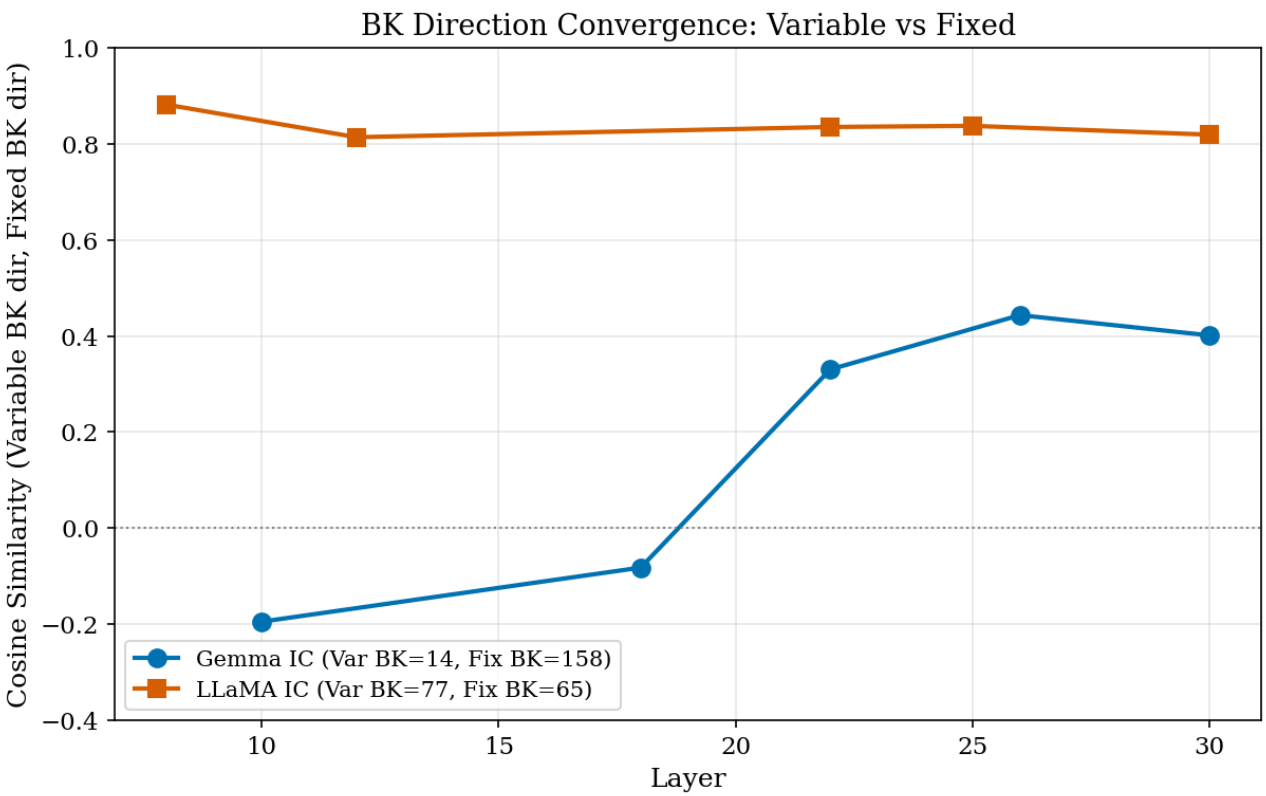

Figure 5: BK direction cosine across layers — Gemma vs LLaMA IC. Gemma (blue): negative at shallow, positive at deep. LLaMA (orange): cos > 0.81 at all layers. Gemma's shallow negative is likely a sample-size artifact (Var BK=14 vs LLaMA's balanced 77 vs 65).

Caveat: Gemma IC Variable BK=14 is extremely small. LLaMA's balanced sample (77 vs 65) provides more reliable estimates. When sample sizes are balanced, BK directions converge at all depths.

4.2 Shared BK Neurons (Interaction Regression)

Per-neuron regression: `activation` $\sim \beta_1 \cdot$`outcome` + $\beta_2 \cdot$`bet_type` + $\beta_3 \cdot$(`outcome` $\times$ `bet_type`). Shared BK neurons: β_1 significant ($p < 0.01$) + β_3 not significant — BK effect is independent of bet-type.

Table 22: Shared BK Neurons Across Models and Paradigms

	Gemma IC L22	LLaMA IC L22	LLaMA SM L22
n_neurons	3,584	4,096	4,096
n_BK	172	142	1,164
Outcome-sig	1,639 (45.7%)	3,150 (76.9%)	1,052 (25.7%)
Bet-type-sig	3,225 (90.0%)	3,742 (91.4%)	3,027 (73.9%)
Interaction-sig	448 (12.5%)	1,605 (39.2%)	1,495 (36.5%)
Shared BK	**1,182 (33.0%)**	**1,501 (36.6%)**	**72 (1.8%)**

Gemma IC 33.0% and LLaMA IC 36.6% — similar proportions of bet-type-independent BK neurons across architectures. LLaMA SM shows only 1.8% shared neurons because SM BK concentrates in Variable games (72.3% vs 0.4%), making "bet-type-independent" BK neurons nearly impossible.

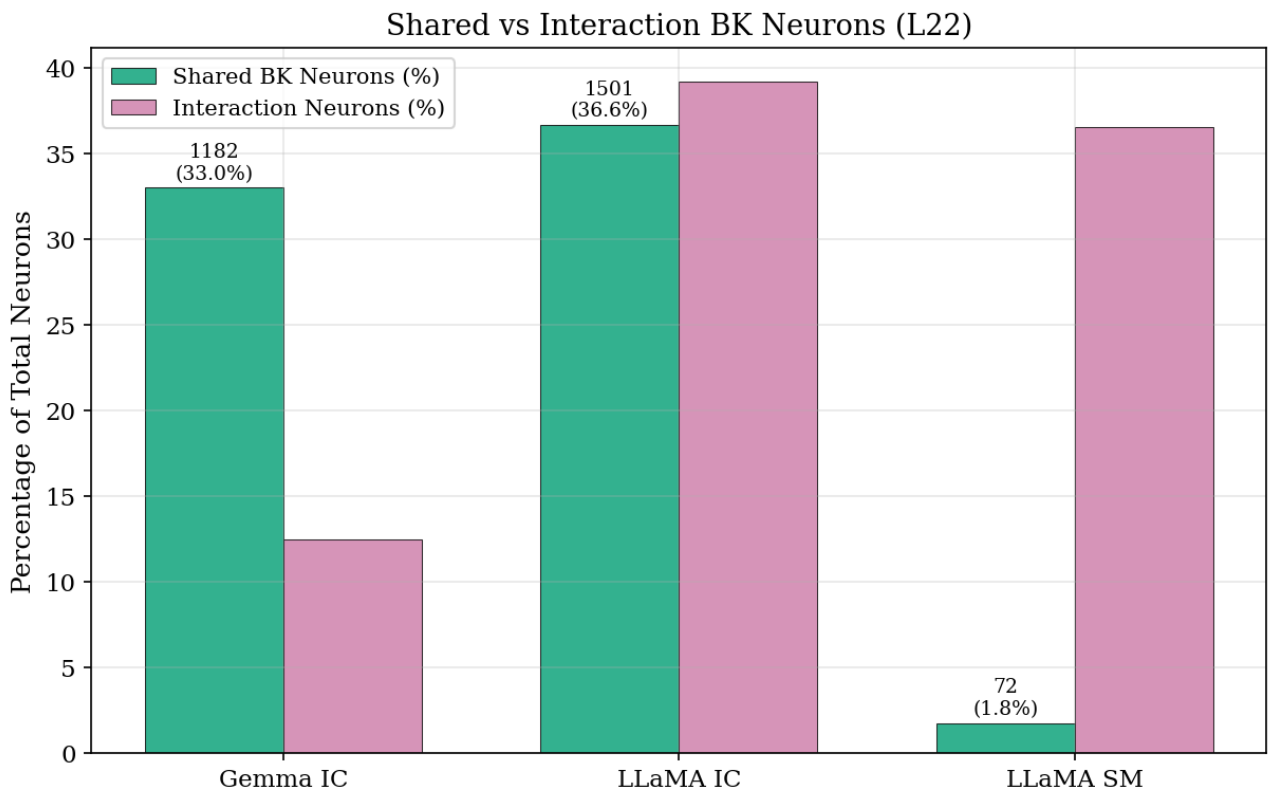

Figure 6: Shared vs Interaction BK neurons. Gemma IC (33.0%) and LLaMA IC (36.6%) show similar proportions. LLaMA SM (1.8%) reflects Variable-dominated BK structure.

4.3 Balance Confound Control

Table 23: Balance Partial Correlation (L22)

	Gemma IC	LLaMA IC
Raw BK-sig neurons	2,578	3,238
Balance-controlled	2,569	3,269
Retained	**99.7%**	**101.0%**

Balance removal preserves 99.7–101% of BK signal. BK representation encodes strategic/behavioral patterns, not simply balance information. LLaMA's 101% indicates balance acts as a suppressor variable.

4.4 Prompt Component Analysis (G/M/W/P/R)

Gemma SM uses hidden state neurons (3,584-dim, 6 layers). **LLaMA SM** uses SAE features (32K, 5 layers). Direct absolute counts are not comparable; percentages are used.

Table 24: Prompt Component BK Rate Effects (Cross-Model)

Component	Gemma SM ratio	LLaMA SM ratio	Cross-model
G (Goal)	**20.75**×	1.26×	Gemma $\gg$ LLaMA
M (Mood/Money)	2.35×	1.31×	Gemma > LLaMA
W (Warning)	2.35×	1.15×	Gemma > LLaMA
H (Hint)	—	1.05×	LLaMA only
R (Reset)	1.72×	—	Gemma only
P (Persona)	1.17×	1.06×	Both negligible

Table 25: Gemma SM Neural Analysis (comp × BK interaction neurons)

Comp.	L10	L18	L22	L26	L30	L33	Pattern
G	109	168	168	308	**552**	**592**	Increases with depth
M	41	23	25	10	8	15	Negligible
R	205	314	235	102	95	97	Mid → decreasing
W	183	70	23	1	0	0	Vanishes with depth
P	53	39	38	8	5	3	Negligible

Table 26: BK-Amplifying Interaction Neurons (Gemma SM)

Comp.	L10	L22	L30	Interpretation
G	0	0	1	G drives BK independently; does not modulate BK
R	18	63	15	R amplifies BK at mid-layers
W	42	3	0	W has shallow-only BK amplification
M, P	2–12	3–6	0	Negligible

Table 27: LLaMA SM Neural Analysis (% of active SAE features)

Comp.	L8 int%	L12 int%	L30 int%	L30 amp%	Pattern
G	45.5%	61.5%	**63.0%**	8.2%	Interaction ↑ depth; low amp
M	**55.4%**	48.2%	**51.2%**	**14.5%**	Highest amp all layers
H	25.9%	31.2%	33.3%	7.4%	Weak
W	36.2%	37.2%	19.2%	5.8%	Shallow-strong, deep-weak
P	26.0%	27.1%	31.2%	4.8%	Weak

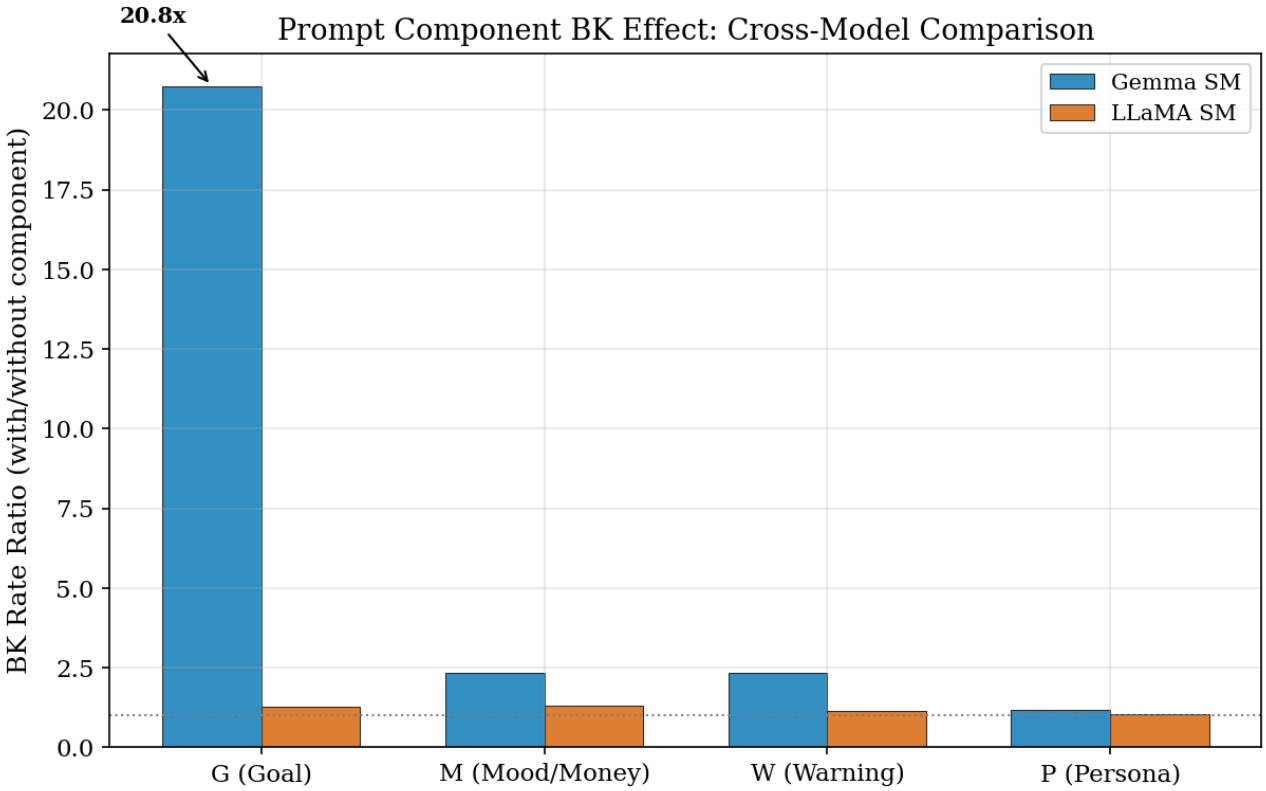

Figure 7: Prompt component BK effect. G(Goal) produces 20.8× BK increase in Gemma (dominant) but only 1.26× in LLaMA. Same prompt, 16× different behavioral effect. M, W, P are weak in both.

G drives BK independently in both models (amplifies_bk ≈ 0 in Gemma, 8.2% in LLaMA). Warning

effects are shallow-only in both models (vanishes at deep layers). M(Money) is LLaMA's highest BK amplifier (14.5% at L30), negligible in Gemma.

4.5 G-Prompt BK Direction Alignment (Gemma L22)

G-prompt direction ($G_{mean} - nonG_{mean}$) vs BK direction ($BK_{mean} - Safe_{mean}$) cosine:

Table 28: G-Prompt Alignment

Paradigm	$\cos(G, BK)$	BK with G	BK without G
IC	−0.143	10.9%	10.6%
SM	**+0.850**	5.2%	0.3%
MW	+0.367	2.5%	0.9%

In SM, G-prompt direction aligns strongly with BK direction (cos = +0.85). G pushes the activation space directly toward BK. This is consistent with G driving BK independently — G moves the hidden state toward BK without modulating existing BK representations.

4.6 Bet Constraint Linear Mapping (Gemma IC L22)

Table 29: Bet Constraint → BK Projection

Constraint	n	Mean BK prob	Actual BK rate
c10 ($1–$10)	400	0.000	0.0%
c30 ($1–$30)	400	0.056	5.2%
c50 ($1–$50)	400	0.212	16.8%
c70 ($1–$70)	400	0.270	21.0%

Linear correlation: $r = 0.979$, $p = 0.021$. (V8: $r = 0.98$, nearly identical.)

4.7 RQ3 Synthesis

RQ3 Evidence Summary

Evidence	Strength	Source
LLaMA IC Var/Fix cos > 0.81 (all layers)	**Strong**	Balanced sample (77 vs 65), §4.1
Shared BK neurons ∼33–37% (both IC)	**Strong**	Cross-model consistent
Balance confound control (both models)	**Strong**	99.7–101% retained
G independent BK driver (both models)	**Strong**	amp_bk≈0 (Gemma), 8.2% (LLaMA)
W shallow pattern (both models)	**Moderate**	Vanishes deep in both
Gemma IC Var/Fix cos convergence	**Moderate**	Deep cos > 0.33 but Var BK=14
G behavioral effect size difference	**Caution**	20.75× vs 1.26×
M amplification difference	**Moderate**	LLaMA M=14.5%, Gemma M≈0

5 Limitations & Critical Assessment

5.1 Permutation-Validated Claims

Table 30: Validated vs. Unvalidated Claims

Claim	Test	Result
Factor decomposition 67–71%	100-perm shuffle	$p = 0.000$ (null 1%) ✓
BK-inhibiting L30	Binomial	All $p < 0.001$ ✓
Classification AUC	100-perm shuffle	$p = 0.000$ ✓
Gemma 3-par. consistency (L18+)	Binomial vs 25%	$p < 5 \times 10^{-3}$ ✓
BK-inhibiting L22	Binomial	LLaMA SM $p = 0.897$ ×
Variable cross-domain d_corr	Bootstrap	mean=−0.075, CI incl. 0 ×

5.2 Cross-Model Effect Size Confound

Table 31: Effect Size Confound

	Gemma SM	LLaMA SM
n_BK	87	1,164
mean $\|d\|$	0.571	0.209
% $d \geq 0.3$	68.8%	23.2%

Gemma's higher $|d|$ likely reflects small-sample inflation ($N = 87 \rightarrow$ noisy, inflated estimates). LLaMA's lower $|d|$ may converge to true effects with larger N. **Cross-model effect size comparison is unreliable.** Classification AUC is a more robust comparison metric.

5.3 Statistical Power Issues

Table 32: Statistical Power Limitations

Analysis	Weakness	Impact
Var BK direction (Gemma IC)	$n_{\text{var_bk}} = 14$	Cosine: single estimate, unknown CI
LLaMA sign-consistency	Chance=50%; 13/16 NS	Weak cross-domain evidence
Gemma L10, L12	Binomial NS ($p > 0.05$)	Shallow signal may be noise

5.4 Multiple Comparison Issues

FDR correction is not systematically applied across all analyses. Expected false positives: Interaction regression (3,584 neurons × t-tests → ∼36 FP at $p < 0.01$); Factor decomposition (581 features → ∼6 FP); Prompt analysis (3,584 × 5 components × 6 layers → many tests). Transfer analyses use permutation tests, partially addressing this. For other analyses, effect size (Cohen's d) should be treated as primary, p-values as secondary.

5.5 Data Constraints

1. **LLaMA MW not extracted**: 3-paradigm cross-model comparison not possible.
2. **Gemma IC Variable BK=14**: Very limited power for bet-type comparisons.
3. **MW Variable BK=4**: Below threshold ($n = 5$), MW bet-type comparison not possible.
4. **LLaMA hidden states**: Only 5 layers extracted (L8,12,22,25,30), not all 32.
5. **No causal validation**: All analyses are correlational. Activation patching needed.

5.6 BK Rate Heterogeneity

Gemma IC Fixed 19.75% vs Variable 1.75%; LLaMA IC Fixed 8.1% vs Variable 9.6% (reversed); Gemma SM 2.7% vs LLaMA SM 36.4% (13×). Cross-model and cross-paradigm comparisons must account for base rate confounds.

5.7 Prompt Component Model-Dependence

G component: Gemma 20.75×, LLaMA 1.26×. "Prompt condition invariance" applies to neural structure (both models show G drives BK independently) but not to behavioral effect size.

6 Next Steps

1. ~~**LLaMA Hidden State Extraction**~~ → **Completed (v9.4)**. Symmetric analyses done.
2. **Activation patching**: Causal validation of top cross-domain features.
3. **LLaMA MW extraction**: Complete 3-paradigm comparison.
4. **IC→SM transfer mechanism**: Investigate layer-specific alignment.
5. **FDR correction**: Apply to all per-feature/per-neuron tests.
6. **G component deep dive**: Explain 20.75× vs 1.26× divergence.
7. **BK rate confound control**: Balanced subsampling for IC-SM re-analysis.

A Supplementary Tables

A.1 Hidden State (600 neurons) vs SAE (744 features)

Table 33: Comparison of Analysis Levels

Level	Cross-domain consistent	FDR?	Binomial sig?
Hidden state (3584-dim, L22)	600 (16.7%)	Yes ($p < 0.01$)	—
SAE features (131K, 7 layers)	744 (total), L18+ sig	No FDR	L18+ $p < 5 \times 10^{-3}$

Hidden state (16.7% universal, FDR-corrected) and SAE (L18+ significant vs chance) consistently support cross-domain BK signal existence.

A.2 Gemma SM Prompt Component Detail

Table 34: Gemma SM Prompt Component BK Rates

Component	BK with	BK without	Ratio
G (Goal)	5.2%	0.2%	**20.75**×
M (Mood)	3.8%	1.6%	2.35×
W (Warning)	3.8%	1.6%	2.35×
R (Reset)	3.4%	2.0%	1.72×
P (Persona)	2.9%	2.5%	1.17×

A.3 LLaMA IC→SM Complete Transfer Profile

L5=0.269(p=1.0), L8=0.417(p=1.0), L10=0.373(p=1.0), L12=0.280(p=1.0), L15=0.522(p=0.02), L20=0.493(p=0.
L25=0.783(p=0.000), L30=0.418(p=1.0).